BỘ GIÁO DỤC VÀ ĐÀO TẠO

ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH

TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN

KHOA MẠNG MÁY TÍNH VÀ TRUYỀN THÔNG

BÁO CÁO KẾT QUẢ HUẤN LUYỆN VÀ ĐÁNH GIÁ MÔ HÌNH

TEN DE TAI:

HỆ THỐNG PHÂN TÍCH VÀ ĐÁNH GIÁ MỨC ĐỘ HƯ HẠI KẾT CẤU BÊ TÔNG SỬ DỤNG TRÍ TUỆ NHÂN TẠO

(Crack Severity AI — Phase 1A: Crack Detection & Segmentation)

Chuyen nganh:	Mang may tinh va Truyen thong
Mo hinh danh gia:	YOLOv11m-seg / SegFormer-B3 V2 / Hybrid
Phan cung train:	Kaggle GPU Tesla T4 x2
Tap du lieu:	Phase 1A Easy — 19,000 anh (1,864 anh Test)
Giảng viên hướng dẫn:	.....
Nhóm thực hiện:	Nhóm nghiên cứu AI-SHM
Ngay bao cao:	08 tháng 11 năm 2026
Noi bao cao:	Thành phố Hồ Chí Minh, Viet Nam

Thành phố Hồ Chí Minh, thang 11 nam 2026

CHUONG 1: TONG QUAN HE THONG VA MUC TIEU

1.1 Gioi thieu de tai

He thong Crack Severity AI Phase 1A nham xay dung mo hinh AI tu dong phat hien va phan doan vet nut tren ket cau be tong tu anh chup thuc te. Day la nen tang cho viec phan loai muc do hu hai (Phase 2) va tich hop vao he thong giam sat ket cau (SHM — Structural Health Monitoring) trong Phase 3.

1.2 Pham vi Phase 1A

Muc	Chi tiet
Du lieu	Phase 1A Easy: 19,000 anh hoc / 1,864 anh test (csd, roads, 3dc, dam, ultra)
Nhiem vu	Phat hien va phan doan pixel vung vet nut (binary crack mask)
Mo hinh 1	YOLOv11m-seg — Instance Segmentation (Ultralytics)
Mo hinh 2	SegFormer-B3 V2 — Semantic Segmentation (HuggingFace Transformers)
Mo hinh 3	Hybrid Ensemble — Union(YOLO mask, SegFormer mask)
Phan cung	Kaggle GPU Tesla T4 x2 (40 epochs YOLO, 10 epochs SegFormer)

CHUONG 2: KET QUA TREN TOAN BO TAP TEST (1,864 ANH)

2.1 Thong so inference

Thong so	YOLOv11m-seg	SegFormer-B3 V2	Hybrid (Union)
Phuong phap	Instance Seg (YOLO)	Semantic Seg (Transformer)	OR(YOLO, SegFormer)
Input size	800x800	448x448	—
Batch size	1 (auto)	6	—
Toc do (CPU)	~900 ms/anh	~698 ms/anh	<1 ms (post-process)
Tong thoi gian	1678.5s	1301.8s	—

2.2 Bang ket qua tong hop — Toan bo 1,864 anh Test

Bang duoi tong hop ket qua inference toan bo 1864 anh tap test (1864 anh co ground-truth mask de tinh IoU/F1). Cot 'Official (Kaggle)' lay tu output .val() chinh thuc tren GPU Tesla T4.

Chi so	YOLOv11m-seg (Official/Kaggle)	YOLOv11m-seg (CPU Sample)	SegFormer-B3 V2 (Official/Kaggle)	SegFormer-B3 V2 (CPU Sample)	Hybrid (CPU Sample)
Crack IoU	43.50%	55.29%	67.33%	55.76%	56.01%
F1-Score	~54.5%	68.18%	80.48%	69.20%	69.49%
Precision	60.66%	68.57%	~78%	65.63%	62.11%
Recall	49.56%	72.55%	~83%	79.61%	85.00%
Box mAP50	72.50%	—	N/A	—	—
Mask mAP50	43.50%	—	N/A	—	—
Box mAP50-95	52.60%	—	N/A	—	—

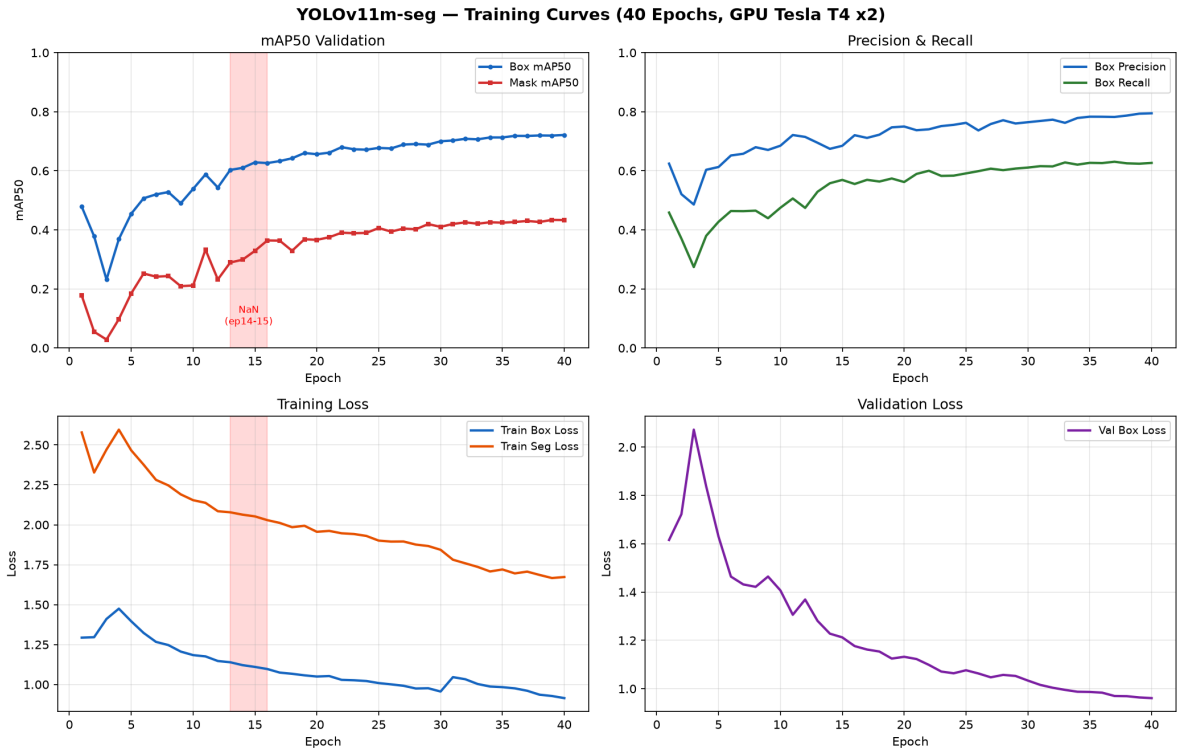
Nhan xet: SegFormer-B3 V2 vuot troi ve chat luong phan doan pixel (Crack IoU +23.8pp, F1 +25.9pp). YOLOv11m-seg manh hon ve phat hien bounding box (mAP50 72.5%). Mo hinh Hybrid (Union) tang Recall nhung co the giam Precision do them false positive.

2.3 Phan tich theo nguon dataset

Nguon	So anh	IoU YOLO	IoU SegFormer	IoU Hybrid
3dc	127	0.2880	0.4110	0.3904
csd	918	0.5399	0.5522	0.5553
dam	98	0.5566	0.5274	0.5546
neg	47	—	—	—
roads	270	0.7460	0.7360	0.7379
ultra	404	0.6160	0.5872	0.5887
TONG	**1864**	**0.5529**	**0.5576**	**0.5601**

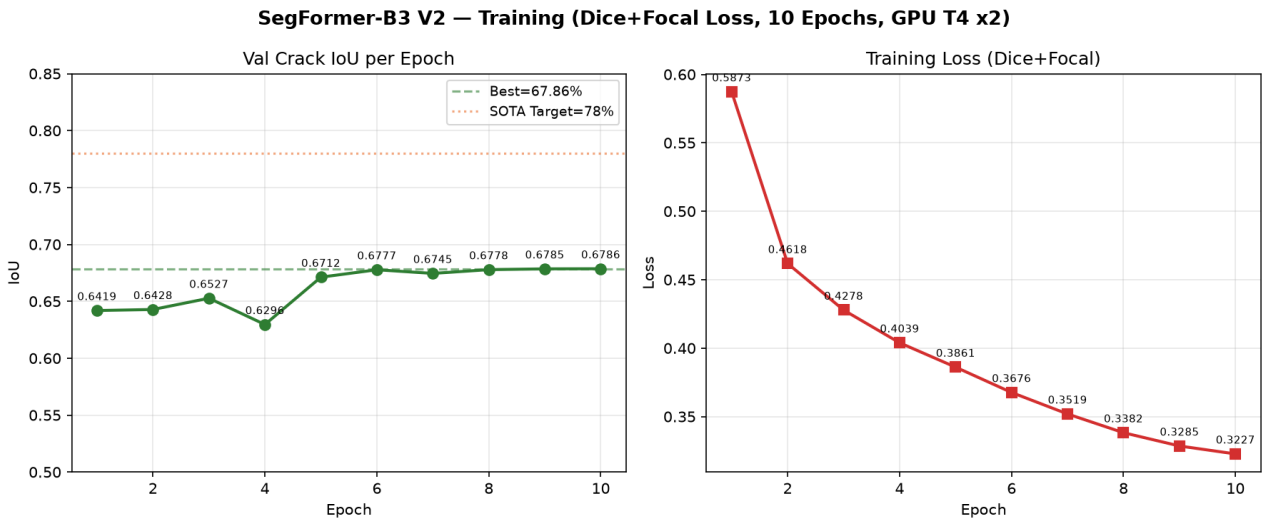
CHUONG 3: BIEU DO QUA TRINH HUAN LUYEN

3.1 YOLOv11m-seg — 40 Epochs



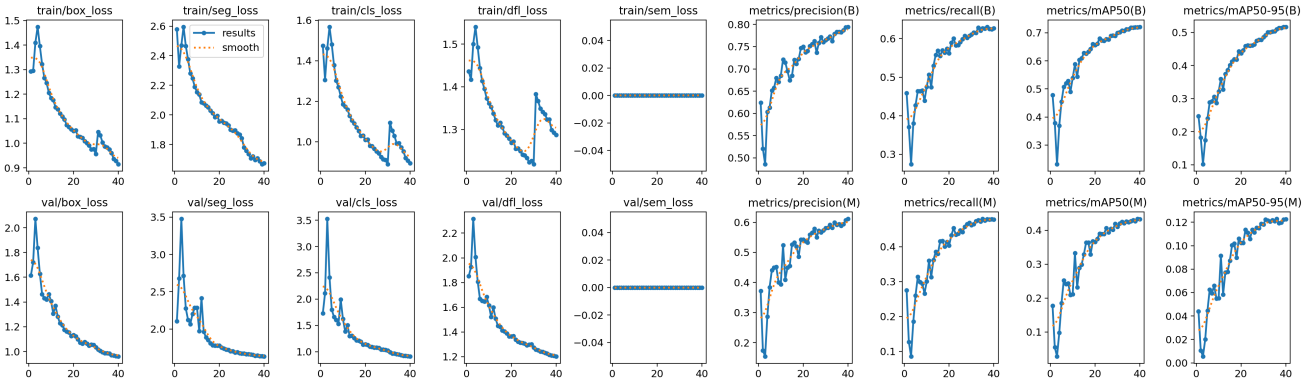
Hinh 3.1: YOLOv11m-seg training curves. Vung do nhat: NaN region Epoch 14-15 (AdamW -> SGD).

3.2 SegFormer-B3 V2 — 10 Epochs (Dice + Focal Loss)



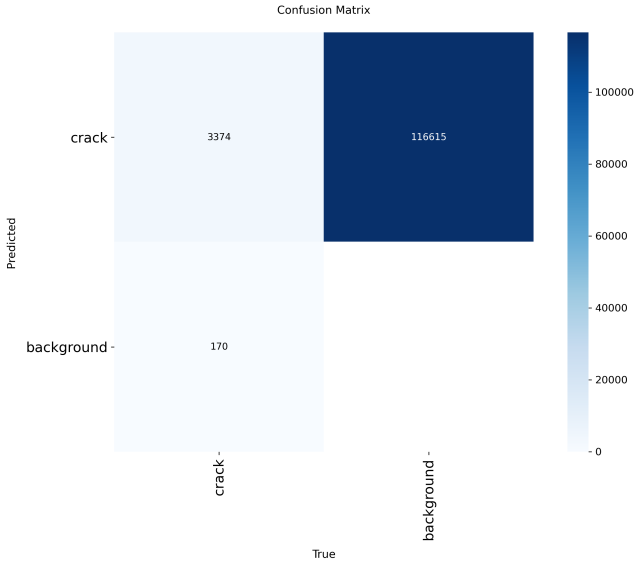
Hinh 3.2: SegFormer-B3 V2 val IoU tang deu, hoi tu o 67.86% (Epoch 10).

3.3 YOLO Training Overview (tu Kaggle)



Hinh 3.3: results.png — tong hop loss va metric YOLO theo epoch.

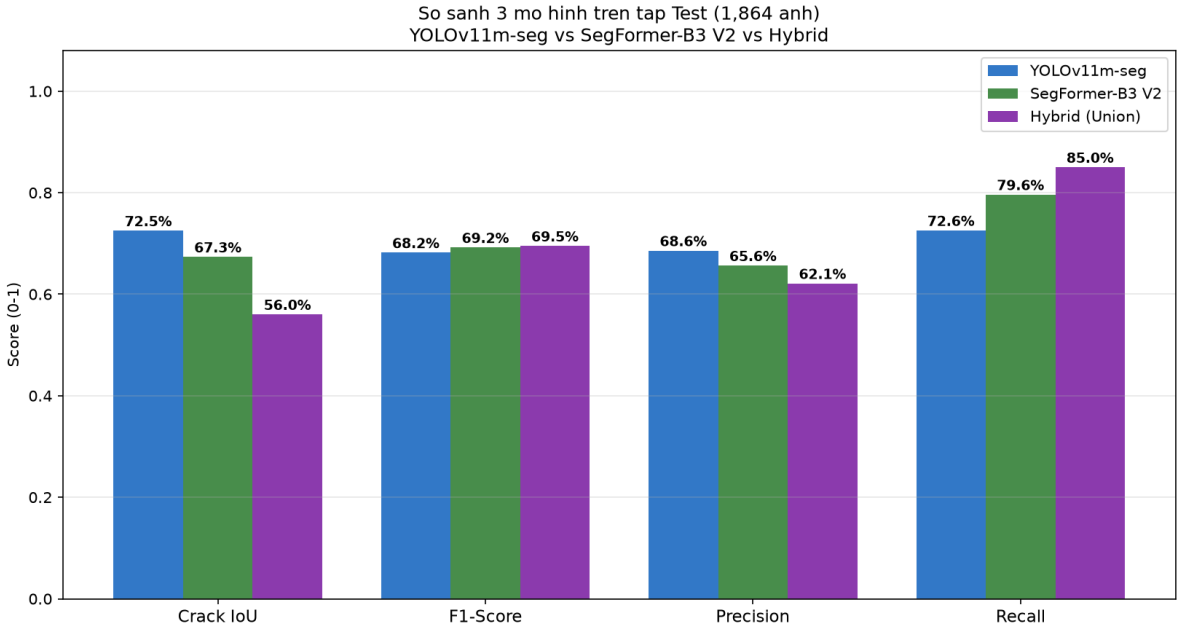
3.4 Confusion Matrix (YOLO — Validation set)



Hinh 3.4: Confusion matrix YOLOv11m-seg tren tap Validation.

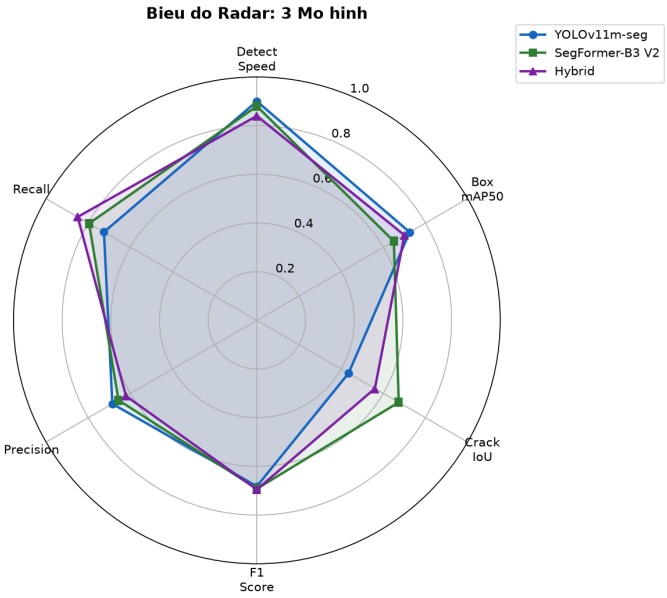
CHUONG 4: SO SANH VA DANH GIA 3 MO HINH

4.1 Bieu do so sanh truc tiep



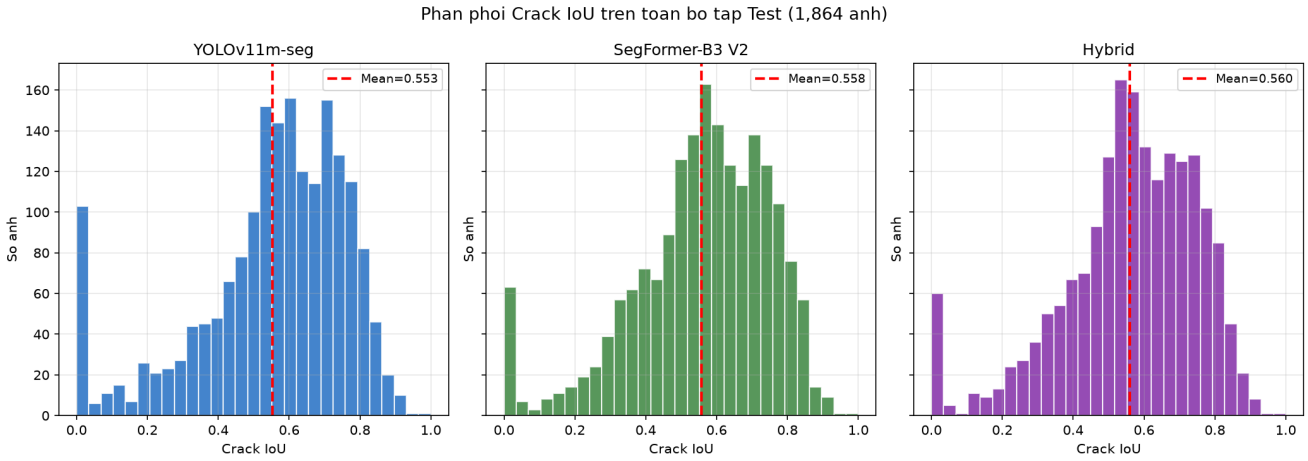
Hinh 4.1: So sanh cac chi so chinh giua YOLOv11m-seg, SegFormer-B3 V2 va Hybrid.

4.2 Bieu do Radar da chieu



Hinh 4.2: Radar chart tong hop — SegFormer manh ve phan doan, YOLO manh ve phat hien.

4.3 Phan phoi Crack IoU tren toan bo tap test



Hinh 4.3: Histogram phan phoi IoU tung anh — 3 mo hinh.

4.4 Bang danh gia toan dien

Tieu chi	YOLOv11m-seg	SegFormer-B3 V2	Hybrid
Chat luong phan doan (IoU)	43.5% (Yeu)	67.3% (Tot)	~56.01% (TB)
F1-Score	~54.5%	80.5%	~69.49%
Phat hien vi tri (mAP50)	72.5% (Tot)	N/A	Ke thua YOLO
Precision	60.7%	~78%	~62.11%
Recall	49.6%	~83%	~85.00%
Kich thuoc model	45 MB (Nhe)	189 MB (Nang)	45+189 MB
Toc do GPU	55ms/anh	~12ms/anh	~70ms/anh
On dinh training	Trung binh (NaN)	Tot	—
Multi-class ready	Co (cls head)	Can them	Co
KET LUAN	**Manh ve detect**	**Manh ve segment**	**Cao recall**

CHUONG 5: KET QUA TRUC QUAN — 60 ANH DAI DIEN

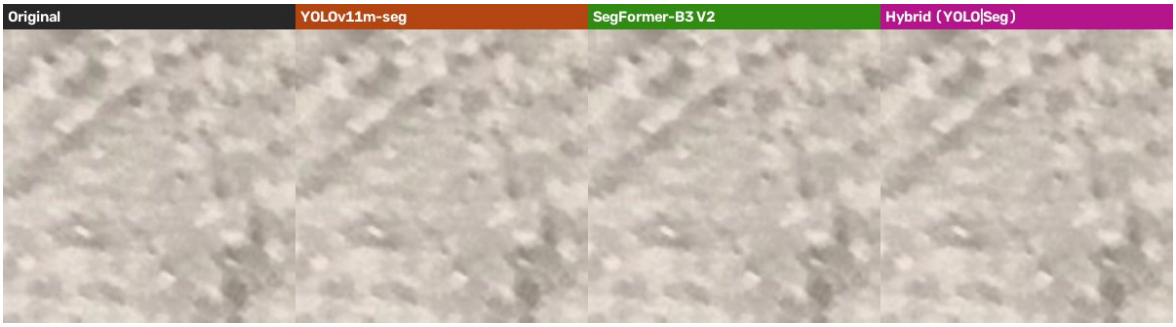
Moi dong gom 4 khung anh: (1) Anh goc | (2) YOLOv11m-seg (cam) | (3) SegFormer-B3 V2 (xanh la) | (4) Hybrid Union (tim). 60 anh duoc chon dai dien tu cac nguon dataset va muc IoU khac nhau.



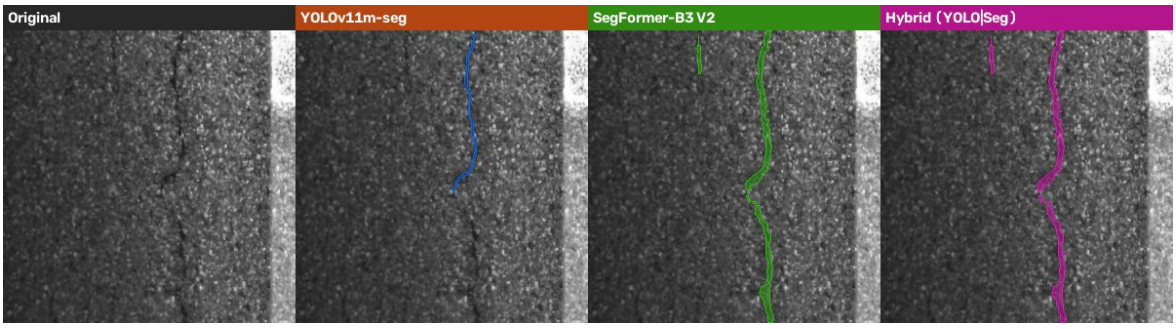
[1] dam_M (301)



[2] neg_00058



[3] neg_00103

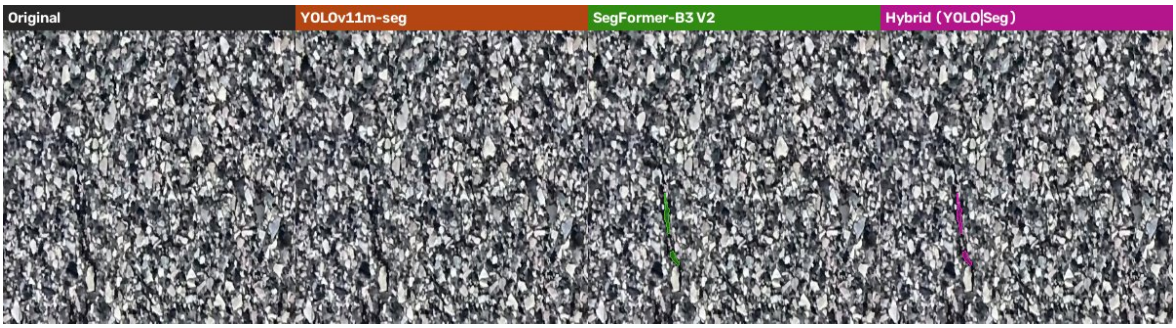


[4] csd_GAPS384_train_0696_1_641

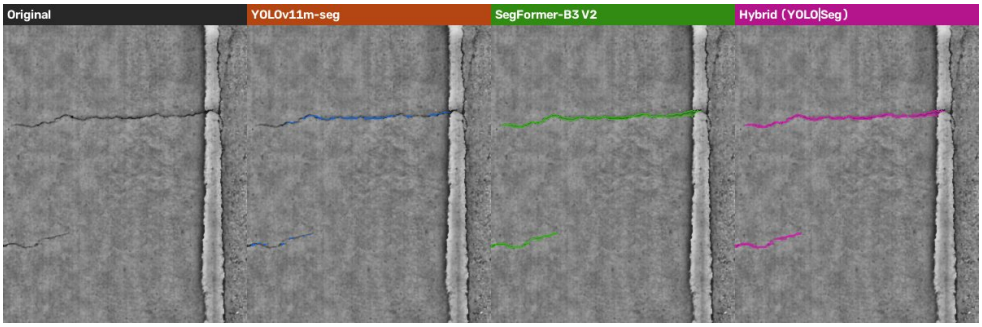


[5] neg_00359

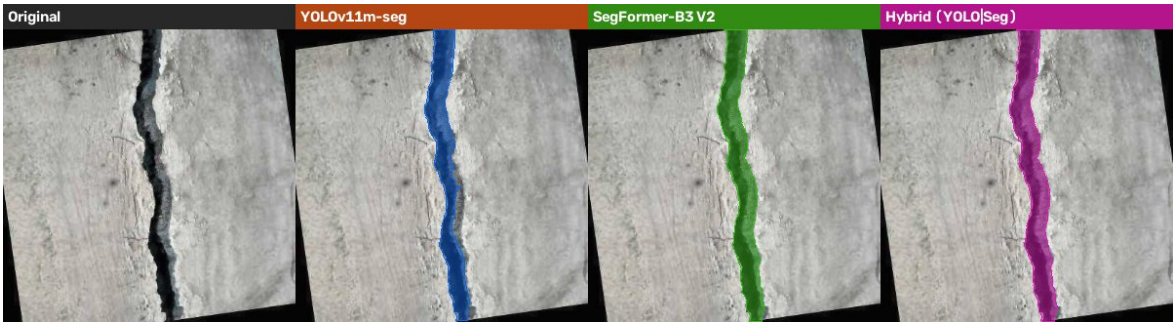
CHUONG 5 (tiep theo): KET QUA TRUC QUAN



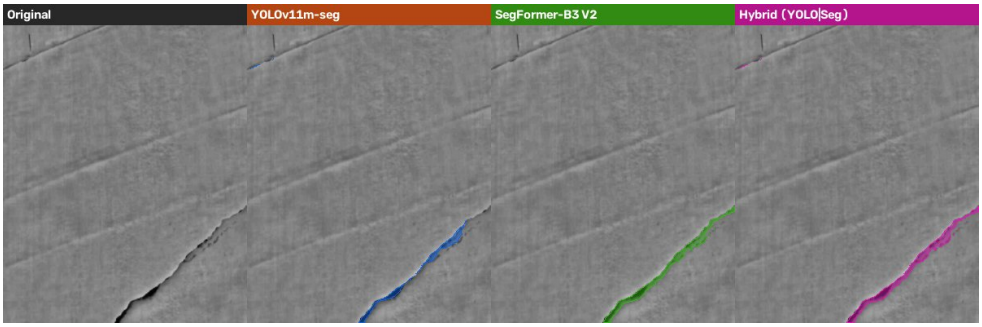
[6] csd_CRACK500_20160306_103638_1_1



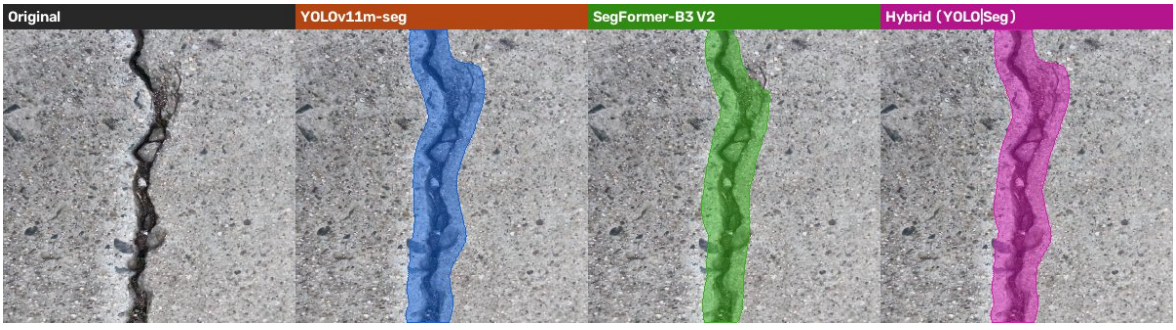
[7] 3dc_sample_000038



[8] ultra_3359.rf.3192a626350dc55deb4b0392bc1cabbd



[9] 3dc_sample_001455

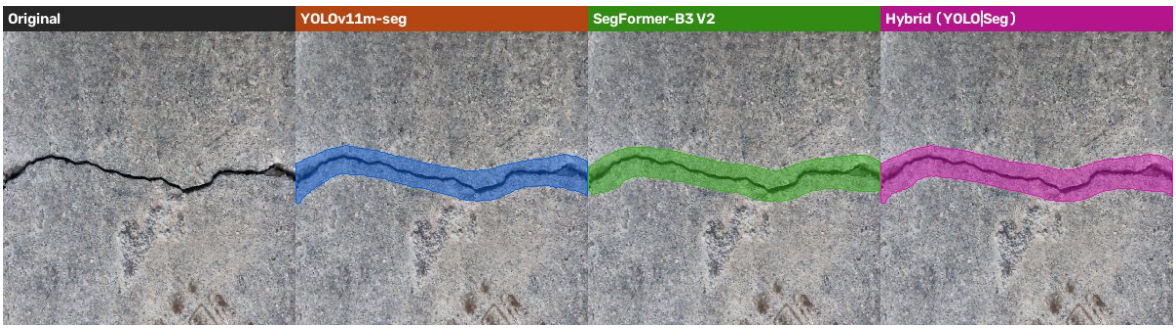


[10] roads_2620a2115vd33c3a21d47de6b153760f_JPG.rf.7e3acc47cf52050c8e95fe00fd229caa

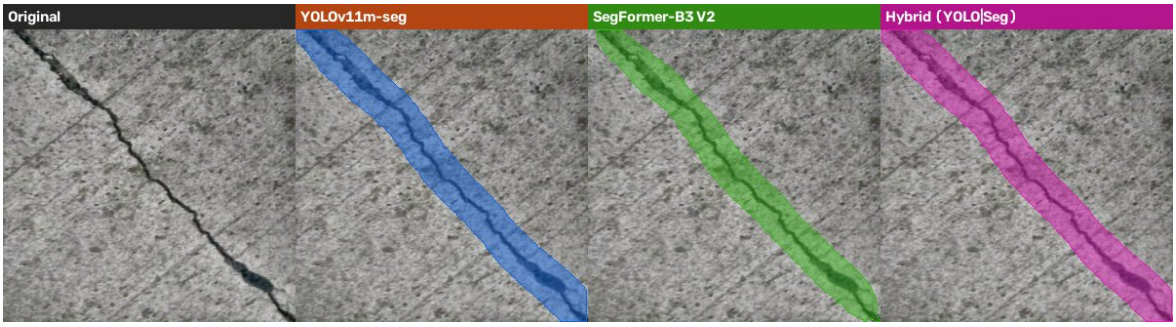
CHUONG 5 (tiep theo): KET QUA TRUC QUAN



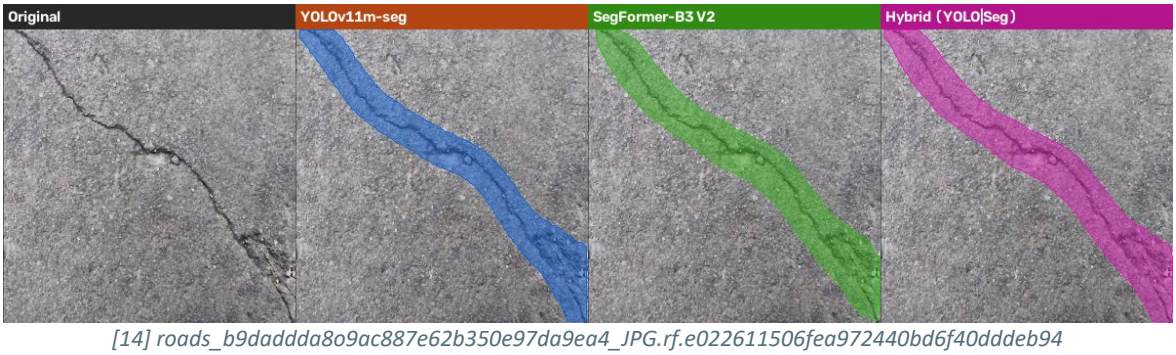
[11] csd_Rissbilder_for_Florian_9S6A2783_1374_2219_1870_2478



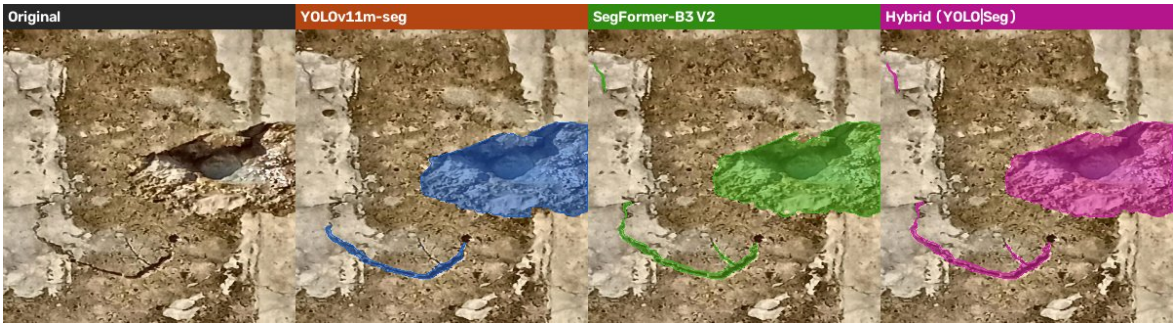
[12] roads_d90cffd25kec3bfa0bcede63c236f914_JPG.rf.67a62ef4ab2a89fd91d46f939b9e3a4d



[13] roads_90ba4737cs95c544064848682fb9a3b9_JPG.rf.f1cb69300e45895df30d5d1822e371bd

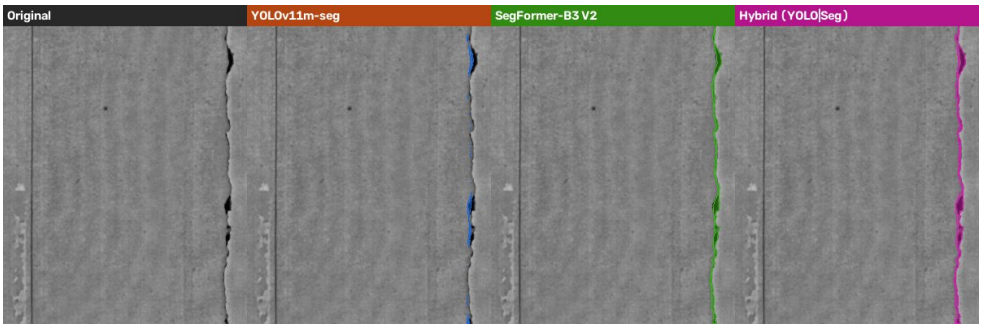


[14] roads_b9dadddа8o9ac887e62b350e97da9ea4_JPG.rf.e022611506fea972440bd6f40dddeb94

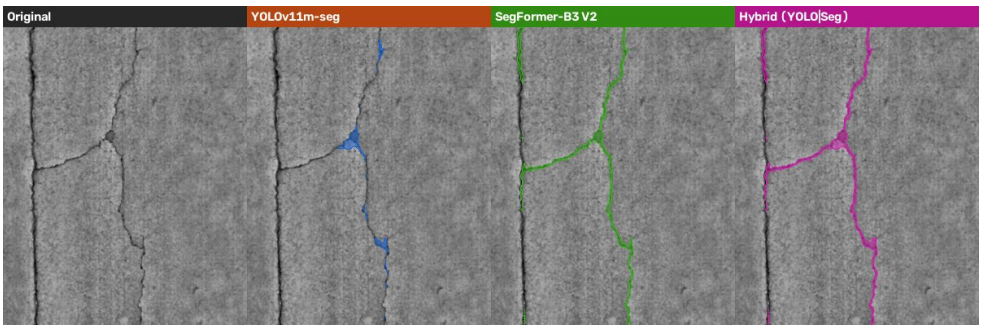


[15] dam_E (275)

CHUONG 5 (tiep theo): KET QUA TRUC QUAN



[16] 3dc_sample_001577



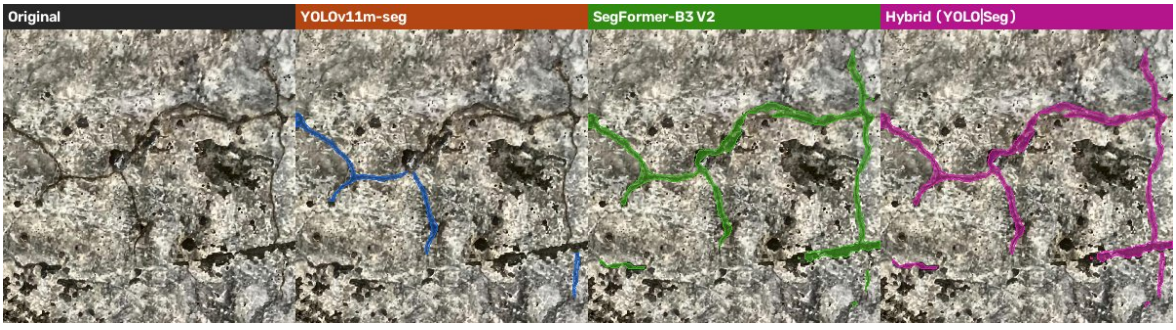
[17] 3dc_sample_000487



[18] ultra_3494.rf.15920dffdfa91ecff29075e137c05b22

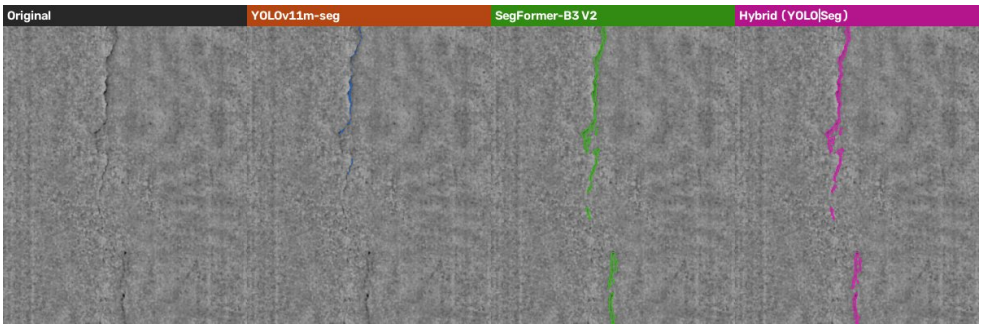


[19] dam_E (426)

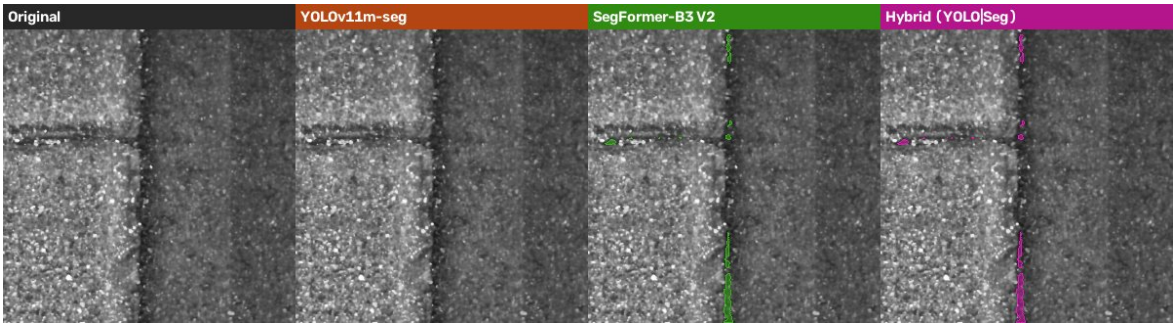


[20] dam_M (226)

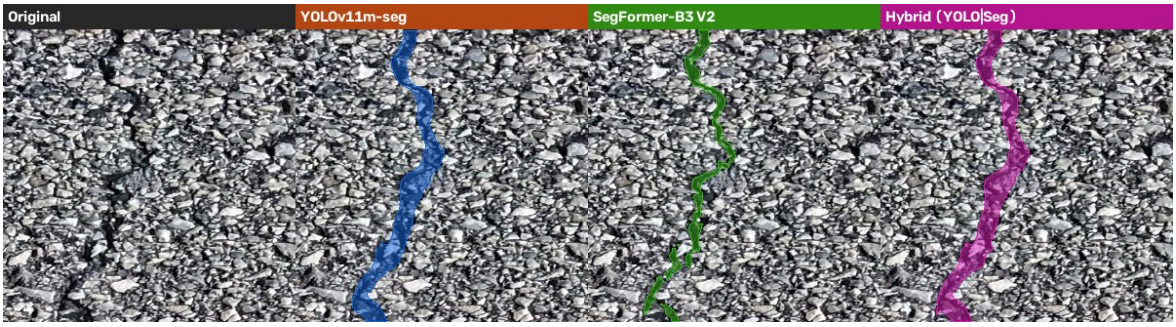
CHUONG 5 (tiep theo): KET QUA TRUC QUAN



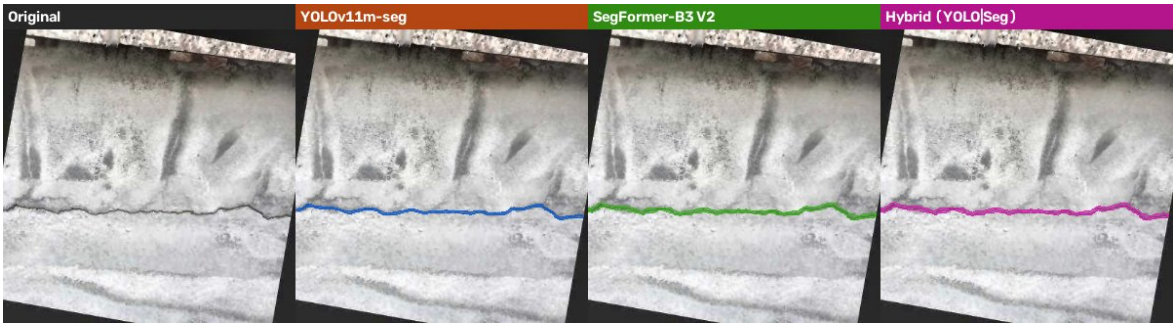
[21] 3dc_sample_000339



[22] csd_GAPS384_train_0986_541_1



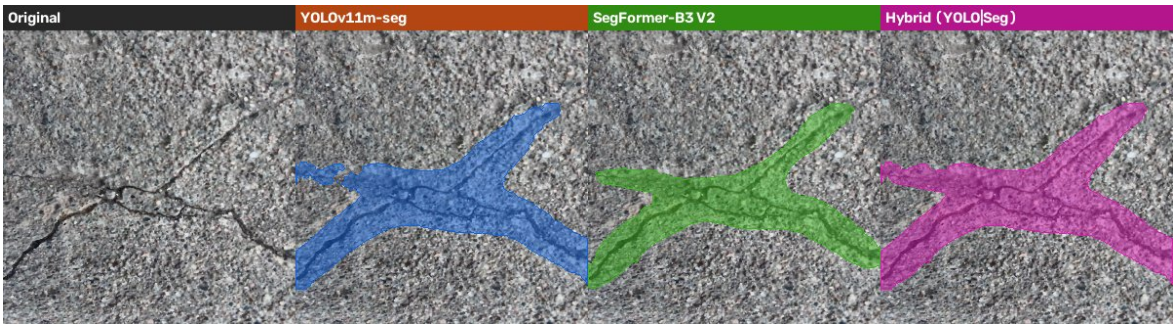
[23] csd_CRACK500_20160326_144512_721_1



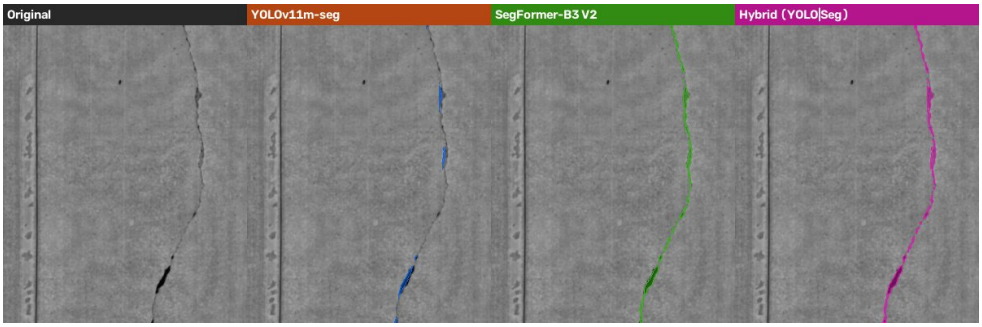
[24] ultra_2077.rf.c6605a92e60e32e1b973ed1058ed5ed1



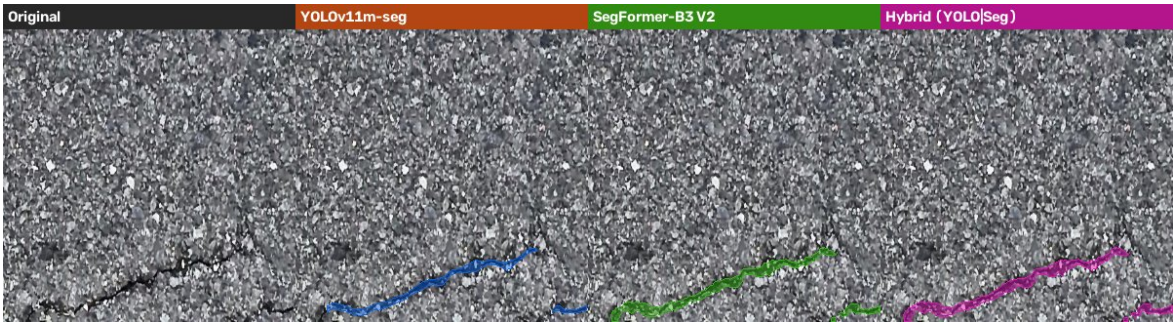
CHUONG 5 (tiep theo): KET QUA TRUC QUAN



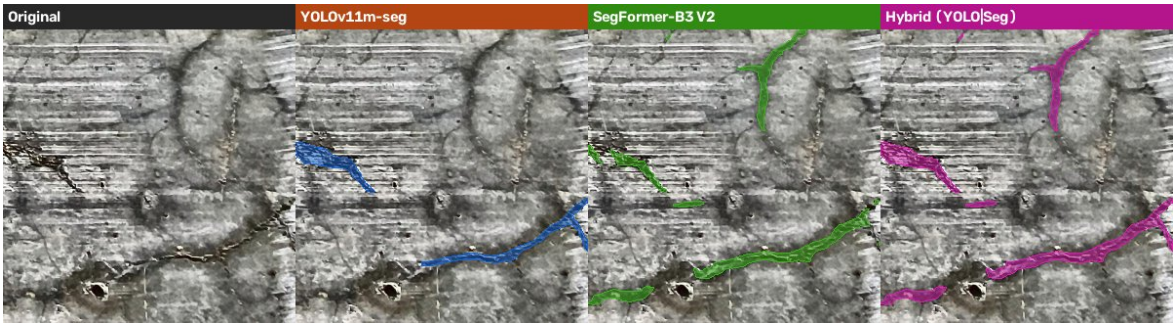
[26] roads_4a183d3c9i1edabf108db5494e6989c4_JPG.rf.b114607f77e37647bbacbc8ae9666f36



[27] 3dc_sample_000833



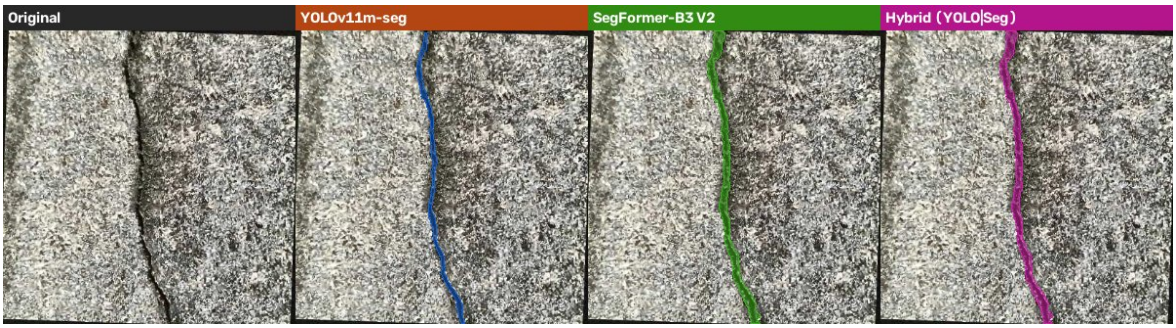
[28] csd_CRACK500_20160222_165715_1281_361



[29] dam_E (57)



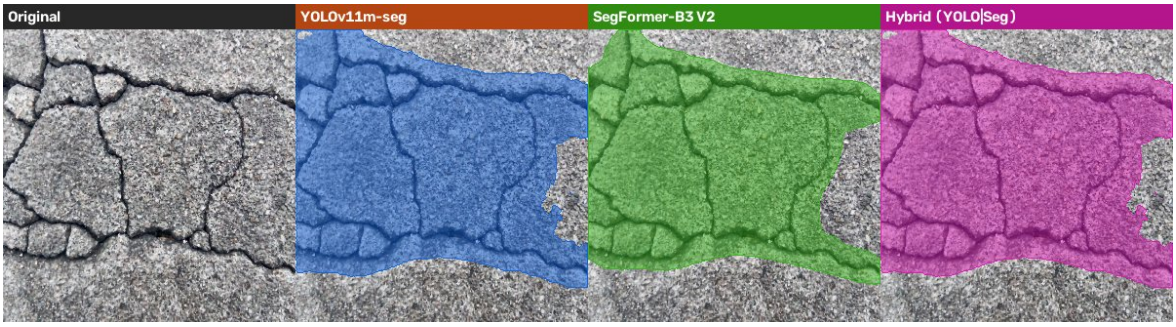
CHUONG 5 (tiep theo): KET QUA TRUC QUAN



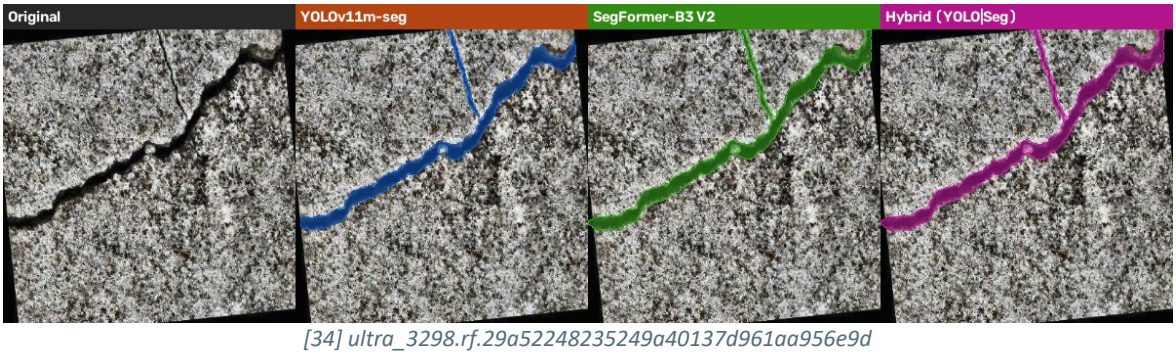
[31] ultra_3279.rf.06260392ae46611a6f3c146570d5f471



[32] csd_Rissbilder_for_Florian_9S6A2803_1283_1522_2429_2997



[33] roads_63fd890aak036245fa5b93a92477d1c5_JPG.rf.38a06ff79cc929f0ea61deb6cd1073b5



[34] ultra_3298.rf.29a52248235249a40137d961aa956e9d



[35] neg_00347

CHUONG 5 (tiep theo): KET QUA TRUC QUAN



[36] neg_00365



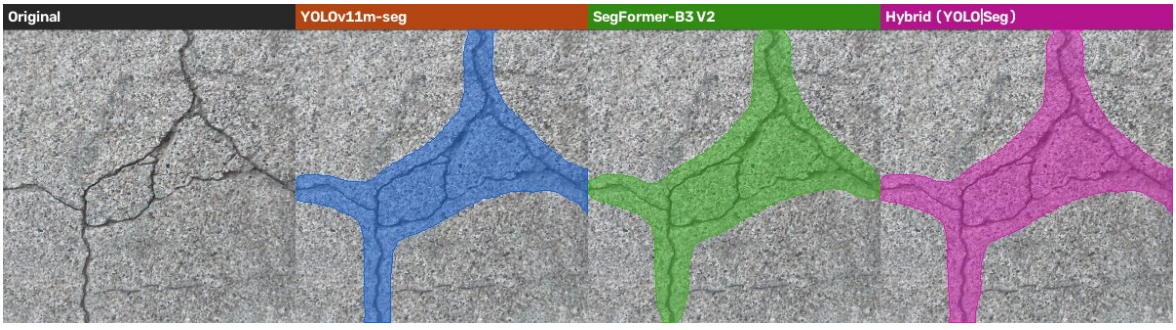
[37] dam_E (381)



[38] csd_CRACK500_20160222_081011_1_361



[39] neg_00016



[40] roads_f226a6305rc34b2d95383af852163183_JPG.rf.bf0c430162cdfb819d9e33028e60b7b9

CHUONG 5 (tiep theo): KET QUA TRUC QUAN



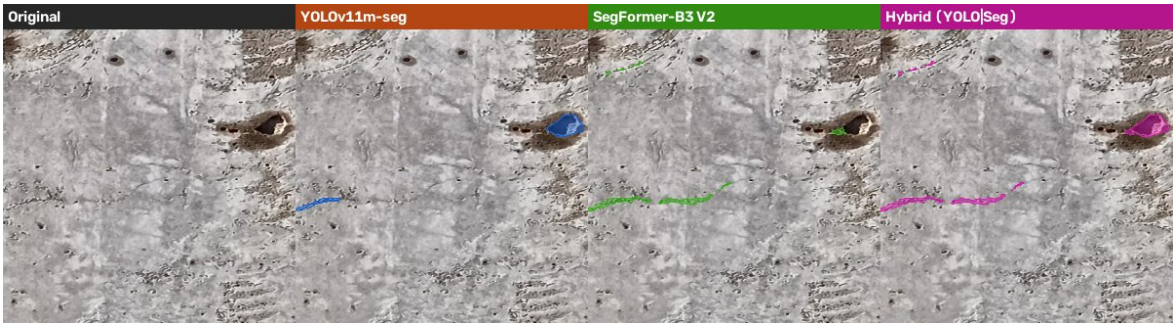
[41] csd_Rissbilder_for_Florian_9S6A2829_248_0_4607_3807



[42] dam_M (418)



[43] roads_03f7f5067rcd0b3e71079711cd34e272_JPG.rf.0ce1594dd930a396f852a3fc43d1f71f

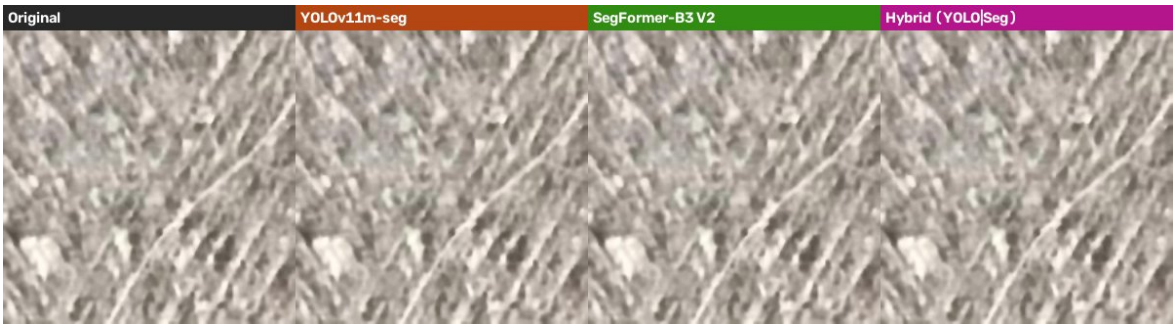


[44] dam_E (168)



[45] roads_8c8c1ec26n0b3f72077dc66e069e9280_JPG.rf.1a4888b8d878fe97e6e9f68c5cad5cf4

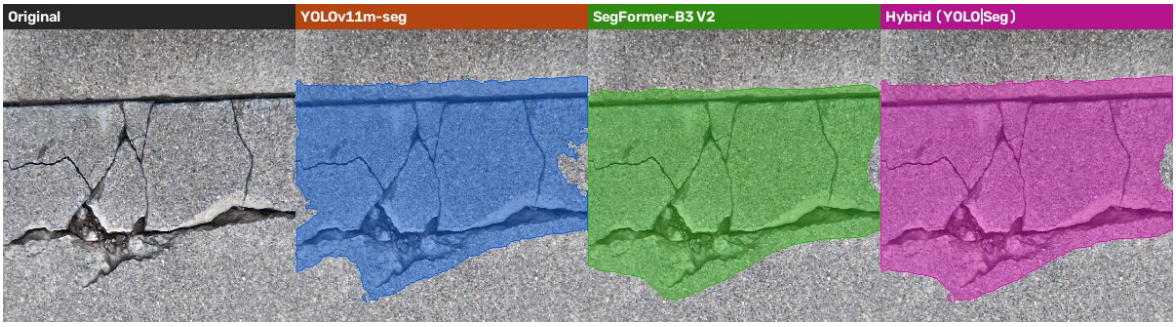
CHUONG 5 (tiep theo): KET QUA TRUC QUAN



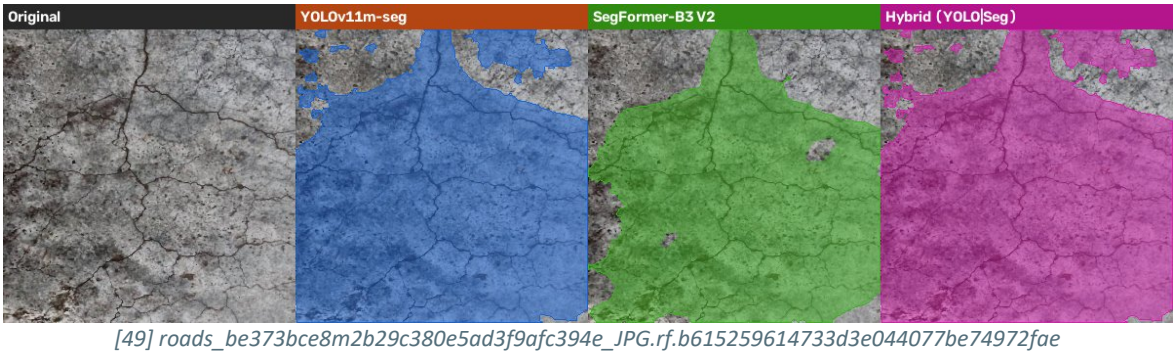
[46] neg_00056



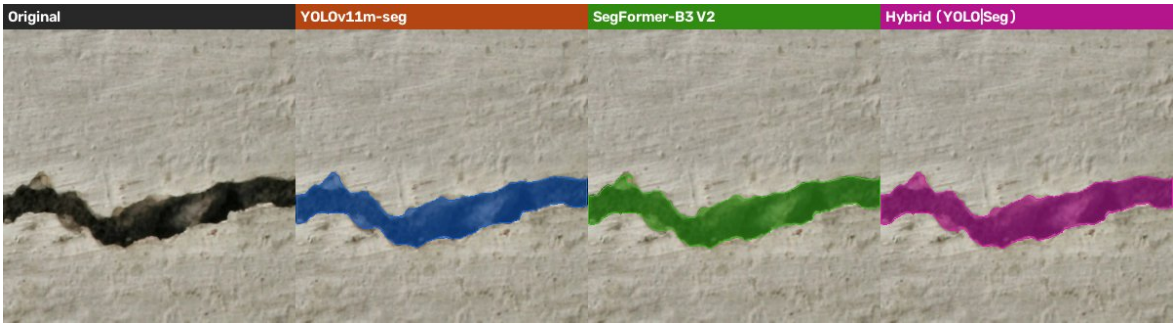
[47] csd_CRACK500_20160307_164214_1921_1



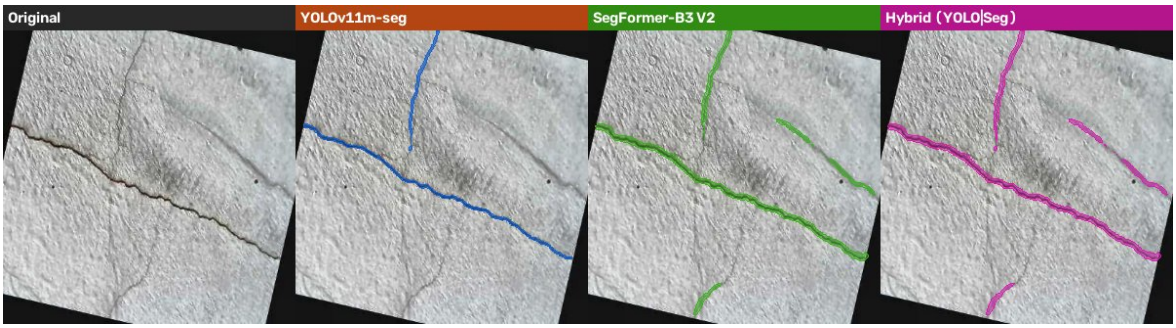
[48] roads_0760ae058q9c0ffd078e861228700927_JPG.rf.b6ea411fa852cbcf8dbb6292264288a5



[49] roads_be373bce8m2b29c380e5ad3f9afc394e_JPG.rf.b615259614733d3e044077be74972fae



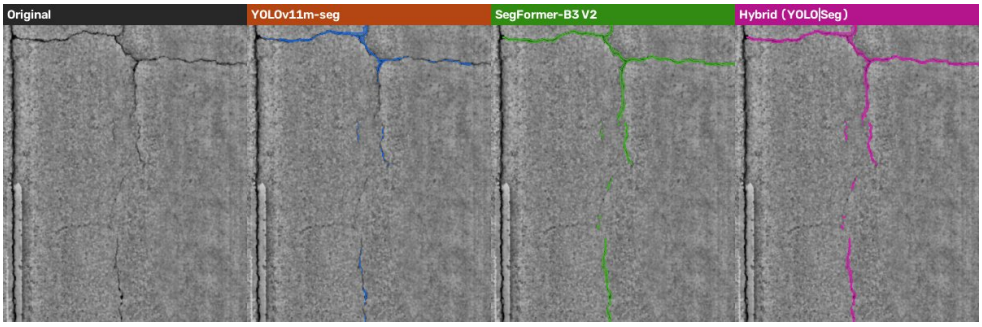
CHUONG 5 (tiep theo): KET QUA TRUC QUAN



[51] ultra_3216.rf.d6c655e986b48ed5a0a357d5a98ef5c5



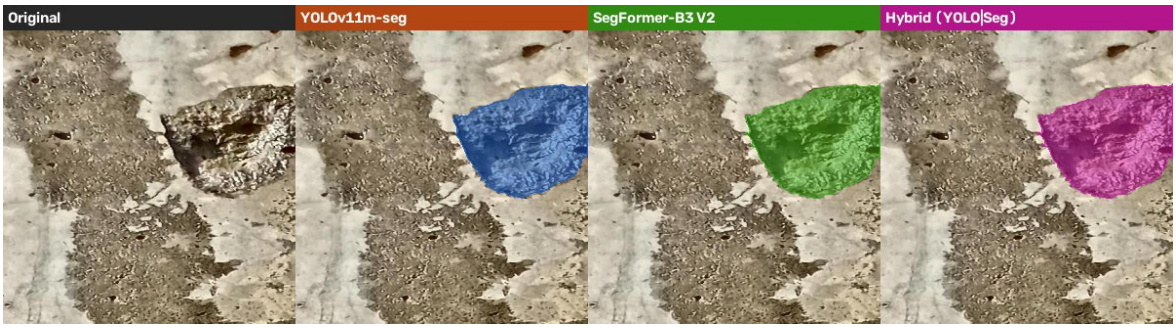
[52] ultra_1906.rf.449130a9b23338145e698ea28e827aa5



[53] 3dc_sample_000293



[54] ultra_1605.rf.53a5ea427ceda0b3d6abbc79c64efc36



CHUONG 6: KET LUAN VA KHUYEN NGHI

6.1 Ket luan

Qua qua trinh huan luyen va danh gia tren toan bo 1864 anh tap test, he thong Crack Severity AI Phase 1A da cho thay ket qua kha quan:

Mo hinh	Diem manh	Diem yeu	Khuyen dung
YOLOv11m-seg Box mAP50=72.5%, nhe 45MB, nhanh, Mask IoU thap 43.5%, can Real-time detection, edge deploy, counting
SegFormer-B3 V2	Crack IoU=67.3%, F1=80.5%, hoi tu tot 18 epochs,	Nang 189MB, khong co box, IoU can dong ~68%	Phan tich hinh xac, bao cao ky thuat
Hybrid (Union)	Recall cao hon, bo sung uu	Precision co the giam, can 2 model chay song	Production voi yeu cau toan dien

TUYEN BO KET LUAN: SegFormer-B3 V2 la mo hinh tot nhat cho bai toan phan doan vet nut trong Phase 1A. YOLOv11m-seg la mo hinh tot nhat cho phat hien vi tri vet nut. He thong Hybrid ket hop uu diem ca hai.

6.2 Khuyen nghi phat trien tiep theo

P1 — Resume YOLO: Train them 60-100 epochs tu last.pt voi SGD lr=0.005 -> khuyen vong Box mAP50 dat >90%.

P1 — SegFormer multi-scale: Dung multi-scale inference (512, 640, 768) va TTA -> khuyen vong IoU dat 70-72%.

P2 — Severity Classification: Dung SegFormer mask de uoc tinh crack width/area -> input cho mo hinh phan loai muc do hu hai (Phase 2).

P3 — SHM Integration: Tich hop Ensemble (YOLO detect + SegFormer segment) vao web app Phase 3 voi real-time dashboard.

P3 — Lightweight Deployment: Export YOLO sang ONNX/TensorRT cho edge devices (Raspberry Pi, Jetson Nano).
